Assignment 3

Reverse outline

Title: Evaluation and comparison of modern forestry inventory methods

Introductory paragraph:
Topic sentence: The airborne remote sensing technologies such as Digital Aerial Photogrammetry (DAP) and Airborne Laser Scanning (ALS) will develop to become leading technologies for modern advanced forest inventory.

This paragraph elaborates about the characteristics of existing different kinds of remote sensing technologies and their limitations from the perspectives of spatial and temporal resolution on application for forest inventory mapping.

Body paragraph 1
Topic sentence: By combining DAP and ALS data all relevant forest inventory attributes can be collected remotely.

This paragraph discusses about the tree attribute details that can be collected through both technologies (DAP and ALS). On the one hand it explains about the limitations on collecting precise information when using only one technology and on the other hand it explains about the completeness of forest species characterization when both the technologies are used as a combination.

Body paragraph 2
Topic sentence: ALS is among the best options for it has achieved the most precise results for all forest inventory attributes.

This paragraph gives details about the feasibility of ALS to apply for different scenarios such as diverse forest vegetation types, the ability to detect lower vegetation levels and to estimate the biomass of deadwood which DAP cannot do.

Body paragraph 3
Topic sentence: DAP is the best option for forest inventorying from the entrepreneurial standpoint where competitive pricing is desirable.
This paragraph highlights the advantage of DAP from the cost effectiveness and the speed of data acquisition.

Conclusion paragraph
Topic sentence: A combination of ALS and DAP is the most viable approach to forest inventory to date.

This paragraph differentiates the two technologies from the points of prediction accuracy and affordability. And it criticizes ALS from the point of data maintenance, shelf life, and temporal data availability. Further this discusses about combining DAP data with ALS data particularly to generate Digital Terrain Models which are essential for working with DAP point clouds.

Peer review

1. Summary of the review

The aim of this paper is to evaluate the modern forest inventory methods, specifically about two emerging remote sensing technologies known as Airborne Laser Scanning (ALS) and Digital Aerial Photogrammetry (DAP). Basically, the paper mention about the available remote sensing technologies at the start and highlights the potential of ALS and DAP in the forest inventory mapping as an emerging technology. Throughout the paper, the author attempts to emphasize the foremost characteristics of ALS and DAP remote sensing technologies that are applicable in accurate forest inventory mapping in the sense of their spatial resolution, prediction accuracy, and from the financial point of view. Furthermore, the author discusses about an array of forest attributes that can be collected through these two technologies and compares them to differentiate the individual level advantages.

In the first paragraph the forest inventory attributes that are obtainable through both technologies in common are presented while dealing with the limitations of ALS technology. And it points out the examples of application of ALS together with DAP deriving the advantage of combining of those two technologies. In the second body paragraph accuracy and preciseness of ALS data are highlighted focusing on the specific characteristics of ALS such as the ability to describe the heterogenic structures, penetrate the outer canopy levels and hence detecting the lower vegetation levels. This paragraph is a commanding attempt in which the author presents views in a logical way. In the third body paragraph, it addresses the pricing factor which is an important consideration in selecting a suitable technology for forest mapping. DAP has been pointed out as the most viable option from the economic standpoint. The conclusion paragraph emphasizes the details presented in the previous paragraphs and stands for combined efforts of ALS with DAP for forest inventorying process.

According to my viewpoint, the overall attempt of shaping the manuscript is outstanding. The author has got a substantial reflective of the subject and has tried to highlight a feasible alternative to the existing technical lapse from the available remote sensing technologies for reliable up to date forest mapping scenario. He has tried to point out a number of sound potential reasons behind his argument. And also, all the references have been listed out accurately. Hence, this paper explicitly presents a good viewpoint in choosing new technologies in forest inventory in the coming era. However, I encountered some challenges in including this paper in the journal due to a few considerations as explained below (Task 2).

I have a background from the forestry and environmental sciences being my major for the bachelor's degree. Hence, I am genuinely interested in the field of application of remote sensing technologies in forest management. I have participated in field level data gathering and mapping exercises in making basic forest inventories and therefore am aware of the attributes of trees and their behaviours that are crucial in forest mapping. In addition to it, I am further interested in remote sensing technologies that are applicable in addressing the different kinds of environmental problems. My current interest in quest is evaluating the possibility of application of the remote sensing technologies in identifying the airborne micro plastics. However, it is quite different from this topic as it needs more precise specific remotely sensed data even though it involves an atmospheric remote sensing component.

2. Evidence and examples

I would like to highlight the following insights about the paper after carefully reviewing the manuscript.

1. **Some paragraphs address one or two subject matters in addition to the topic statement.**

   Example 1: Paragraph 3 explains about two main factors, one is about the cost factor, the other one is about the requirement of a digital elevation model for the use of DAP.

   Example 2: Concluding paragraph explains about several factors such as shelf life of ALS data, unavailability of temporal data with ALS, and about digital elevation model requirement of DAP in addition to the summary of the earlier explained content.

2. **Content gaps/Missing information**

   Examples 1: In the introductory paragraph, it talks about the need of detailed and accurate information for sustainable forest management, but the factor that of what kind of specific data needed is missing. Because information could be gathered from the ground level sources too, other than the remotely sensed data.

   Example 2: Basically, the paper discusses about two remote sensing technologies. However, in the introduction paragraph, a necessary introduction to those two technologies is missing.

   Example 3: In the body paragraph 1, in addition to the achievable results in common, only the limitations of ALS are discussed, but either no capabilities or limitations of DAP are touched on acquisition of specific data on forest attributes. However, at the end, this paragraph discusses about the advantages of combined attempts.

3. **The structure of the paragraphs is deviated from the expected**. The topic paragraphs have been further divided into three or more parts.

   Example 1: The introductory paragraph has been divided in to three parts instead of one continuous paragraph which is not expected in this kind of paper. And the first part and the last part only contain one sentence each.

   Example 2: The concluding paragraph has been divided into four parts in which each part describes a different perspective. This makes the concluding paragraph more debatable.

4. **Repeated statements**

   Example: Body paragraph 3 states about the Digital Elevation Model requirement and it is again explained in the concluding paragraph with more insights rather than giving a brief explanation about the fact.

Suggested major improvements:

1. I would like to suggest combining the body paragraph 1 and 2 together with some more insights as highlighted above to address the main characteristics of those two technologies such as ability to collect data relevant to different forest attributes, their accuracy levels in different aspects.

2. Next, I would suggest you make the third body paragraph to include the limitations of each technology focusing on the single application circumstances of each such as shorter shelf life and unavailability of temporal data sets in ALS and Digital Elevation model requirement of DAP. Here you can discuss the appropriateness of combined efforts. This would comprehend the essay justifying all main points that the author wants to convey through a logical framework.

3. However, there is a contrastive viewpoint emerged from the body paragraph 2 and from the concluding paragraph, i.e. “the discrepancy in prediction accuracy is negligible from a forest management standpoint”, which refers to the DAP context. However, the high accuracy of ALS with compared to DAP is the main point highlighted in this paragraph. Further it is stated that “ALS remains more suitable for diverse forests and forests with high use intensity, for it is better suited to describe heterogenic structures” which direct the reader towards a more specific framework in ALS application circumstances.
   Therefore, more importantly, I would like to highlight the significance of addressing a more specific application scenario within the forest management perspective for the combined application of ALS and DAP rather than addressing the general context.
   For example: Evaluation of combined efforts of ALS and DAP for forest canopy inventorying

Required minor improvements:

1. Thus, it is worthy to include one or two introductory sentences explaining what ALS and DAP do mean. It will help the reader to get a fundamental idea about those technologies at the start.
2. It is important to address the capabilities and limitations of DAP as well before pointing out the combination efforts of both.
3. It is important to state the citations in the introductory paragraph as the citations are missing there.